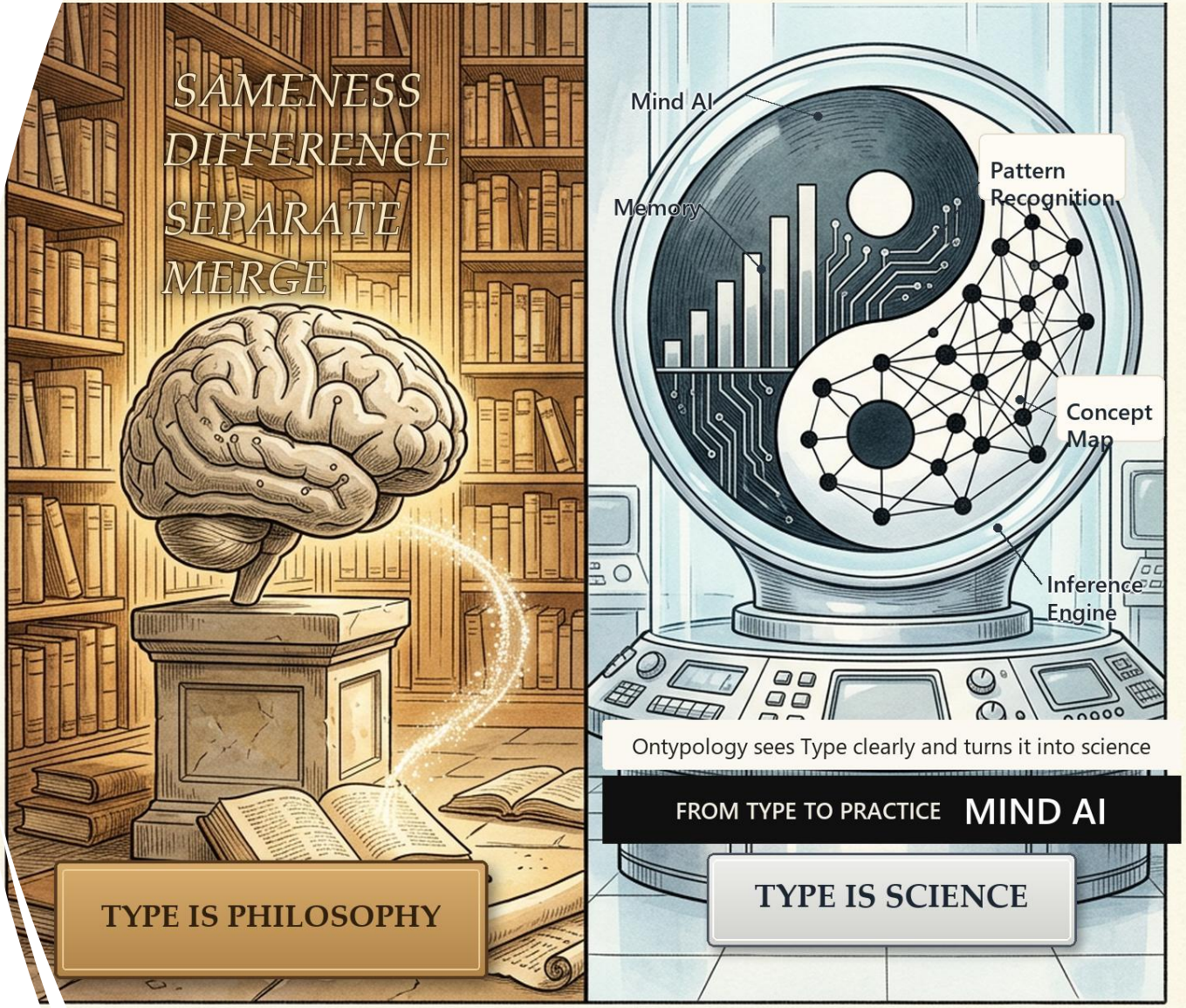
Ontypology

- From sameness and difference to separation and merge, from separation and merge to Type, from Type to understanding—the brain spent eons walking this path.
- A century ago, logic was philosophy; once seen clearly, it became a science.
- Today, exploring “Type” (a concept of the brain) is philosophy. Ontypology sees “Type” clearly, turns it into a science—and practices it as Mind AI.

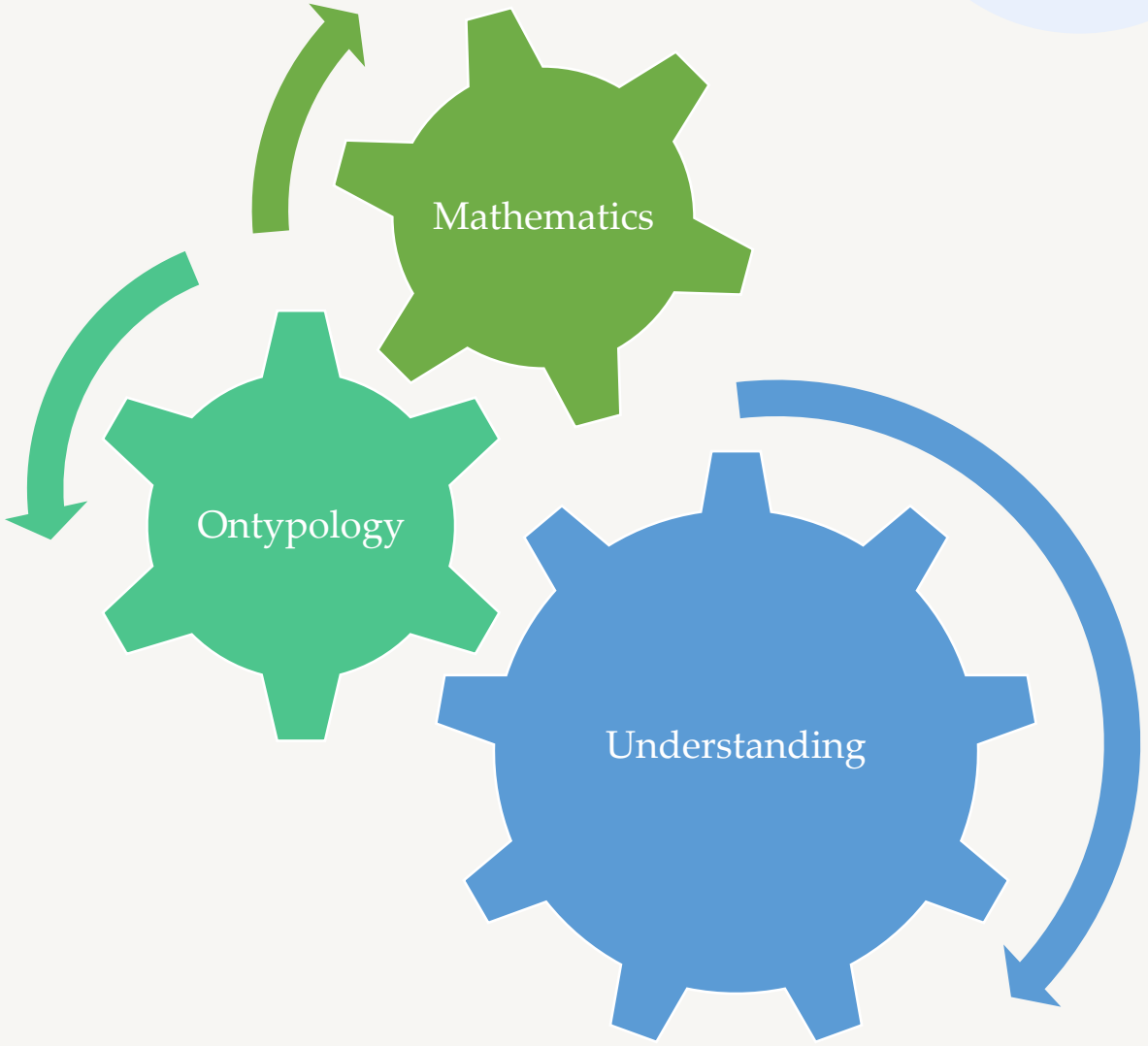
Ontypology ★

Sameness / Difference → Separate & Merge
Type → Understanding → Science



Fei-Fei Li: The Digital Cambrian Explosion

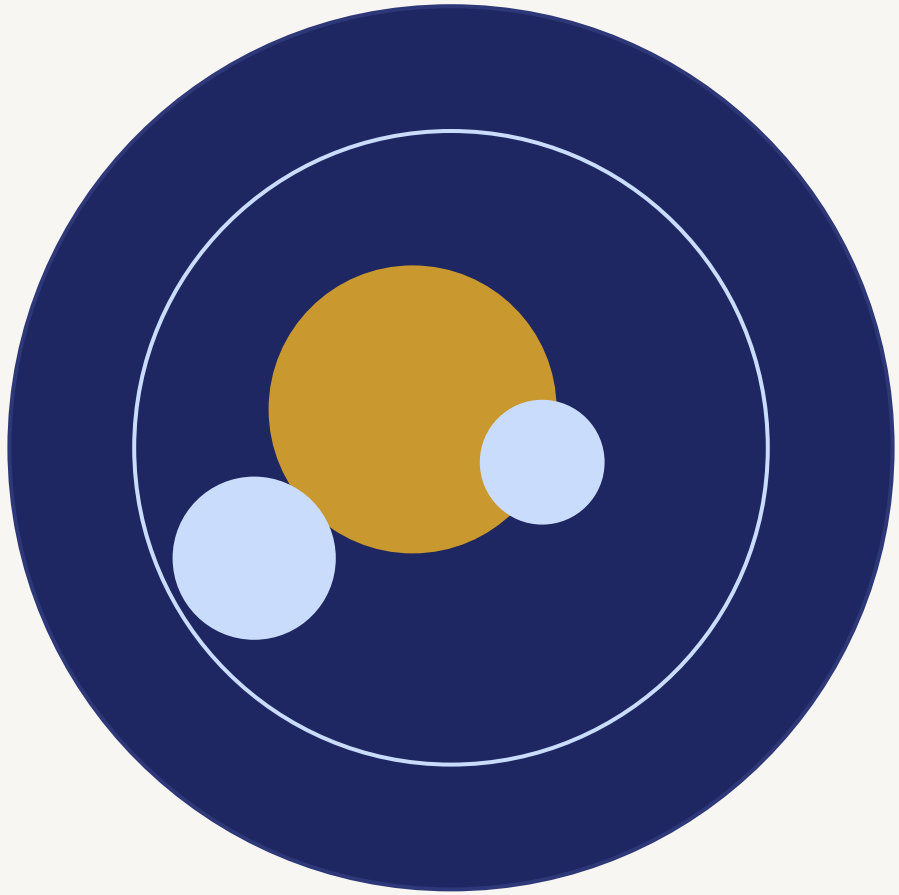
Fei-Fei Li: The Digital Cambrian Explosion
She described how, in the distant Cambrian era, everything was pitch black. A trilobite-like creature first sensed light; sensing light led to a nervous system, then to "**understanding thought**" and finally execution. This is the law of biological development.



Ontypology

Type & Number

First Type, then Number.



From a single sensor in a one-dimensional world, derive the birth order of Type and Number

CORE PROPOSITIONS

Two sentences that run through the whole chapter

Merge

Where there is merge, there is Type

Enclosing originally separate things into one whole—this act itself creates a new, higher-level Type.

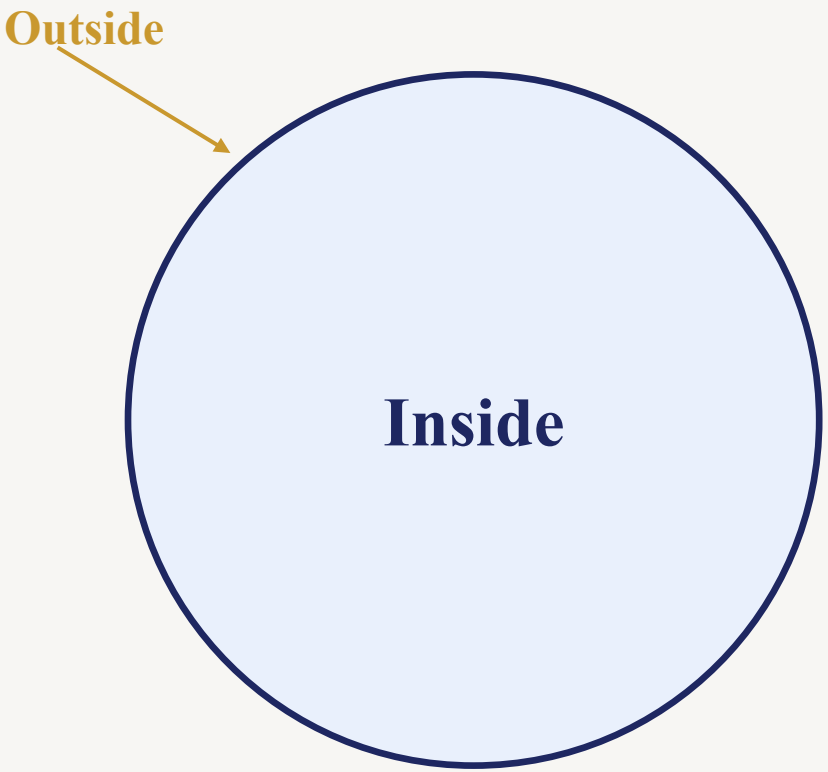
Number

Number = the small Types within a Type, counted and compared

Under one Type, count how many times its Sub-Types appear—Number always depends on a Type that has already been recognized.

THE STARTING POINT OF SPATIAL TYPE

Inside/outside separation is the most basic Type



A circle, separating inside from outside, is a basic Type—it exists, has a boundary, and can be recognized as “this is one thing (a whole).”

Inside/Outside · Group Merge · Entity Merge · Basic Unit

THOUGHT EXPERIMENT

A Single Sensor in a One-Dimensional World

Imagine a one-dimensional world with only one sensor—it senses color once per second, with no other function.

Case A Always the same color



No difference → no space, no time, no Type

Case B The color begins to alternate



Difference appears → space, time, and Pattern are born together

THE BIRTH OF TYPE

The judgment of sameness/difference is the first "Type"



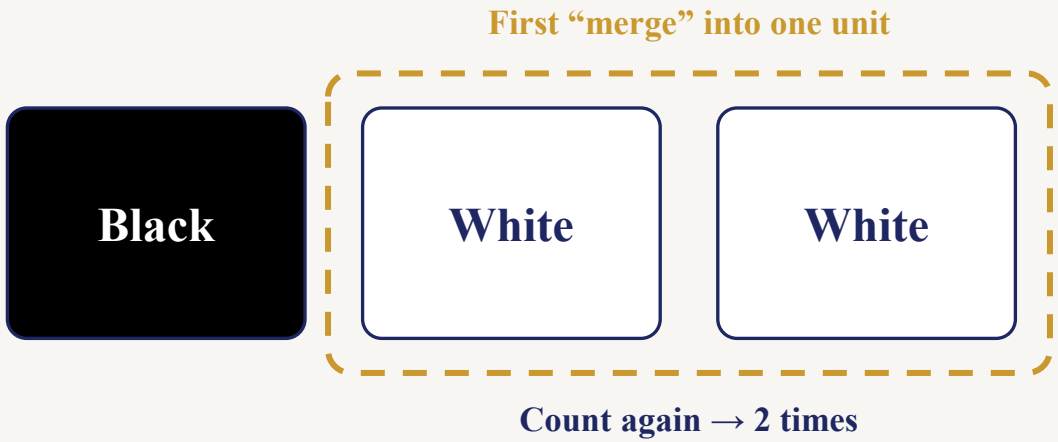
THE MOST PRIMORDIAL ACT OF RECOGNITION

The first "Type" is essentially a Boolean value: "present" or "absent," "is" or "is not," "different" or "same."

It does not yet involve "quantity" or "how many"—it only recognizes that "this is an independent state."

THE BIRTH OF NUMBER

White appears twice—but only after “merge”



Number is not just the act of counting; it presupposes that certain things can be treated as one unit—and that judgment is what “merge” quietly does behind the scenes. If every white were treated as something entirely different, there would be only difference, no sameness, and no way to count.

DEFINING THE CONTRAST

Type answers “what is it?”; Number answers “how many?”

Type

What it is / what it is not

The discernment of sameness and difference.

Without Type, one cannot recognize “this is an independent state.” It is a more primordial, more fundamental cognitive act—

Type can exist without Number.

Number

How many / how many times

A further measurement built on Type.

Only after a Type can be recognized can one count “how many times this Type appears”—

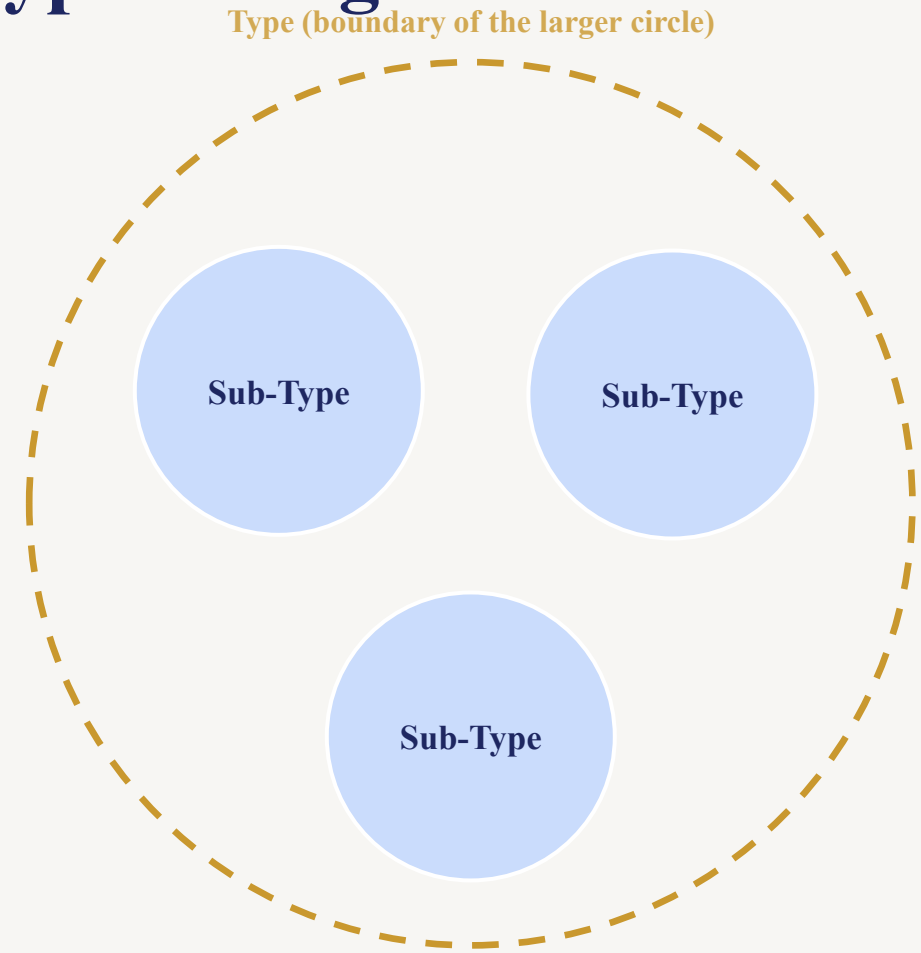
Number cannot exist apart from Type.

A (black) B (black-white) C (white) — sameness and difference let us distinguish these Types; but asking “how many times does A appear?” crosses from Type into Number.

Type is the starting point of the next abstraction: A, B, C → a higher-level abstract Type X

MERGING & STACKING WHERE THERE IS MERGE, THERE IS TYPE

Three circles merge into a group, and a new level of Type emerges



Before merging, **there are only three independent basic Types;**

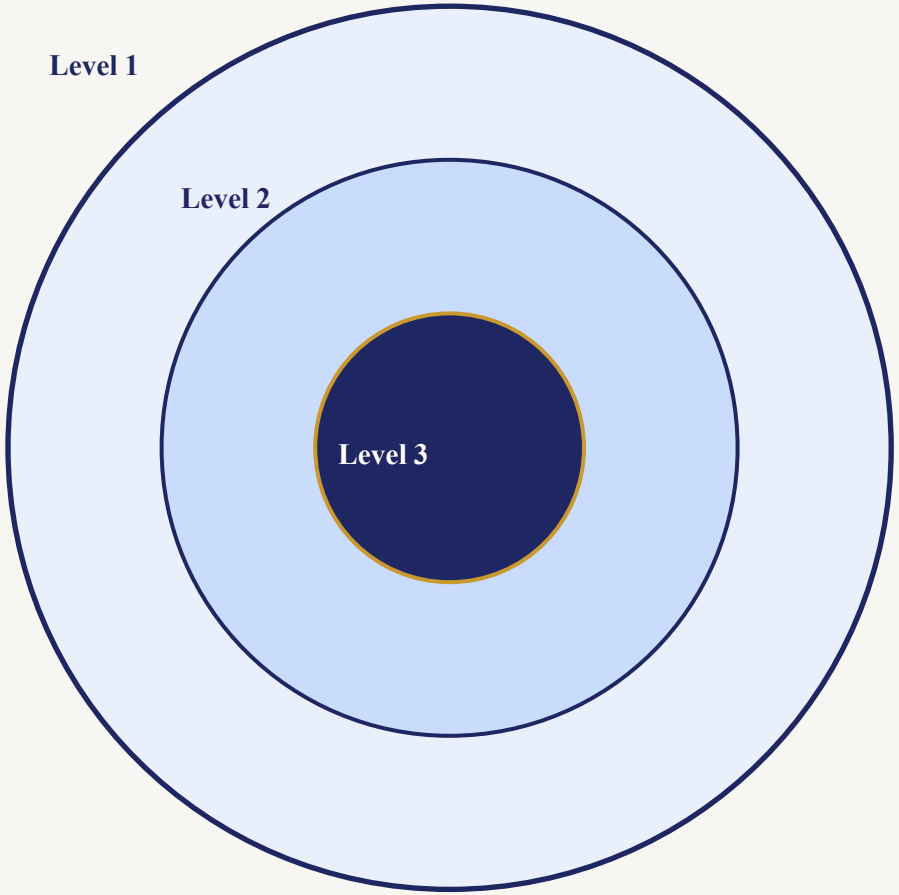
after merging, a new Type appears at a higher level.

- The largest circle (the boundary enclosing the others) → called the “Type”
- Each small circle enclosed within → called a “Sub-Type”

“A Type contains Sub-Types” still answers “what is it?”—a relation of belonging, not “how many.”

THE STRUCTURE OF INFINITE STACKING

Type can stack downward without end; Number is asked anew at every level



The stacking of Types = structural extension

At every level, asking "how many?" generates Number anew

- **Alternating black and white:** first distinguish black from white (Type), then count how many blacks and whites (Number)
- **Type and Sub-Type:** first determine which small circles belong to the large Type (Type), then count how many Sub-Types it contains (Number)

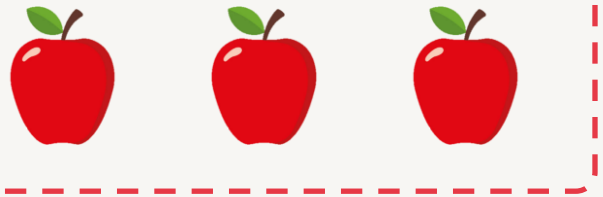
Common ground: recognition always precedes counting—only after "one" and "another" can be separated can one count "how many in total."

CASE STUDY

3 apples, 2 potatoes, 2 peaches

How many fruits are there in total?

Fruit



Apples ×3



Potatoes ×2

Fruit



Peaches ×2

$$3 + 2 = 5$$

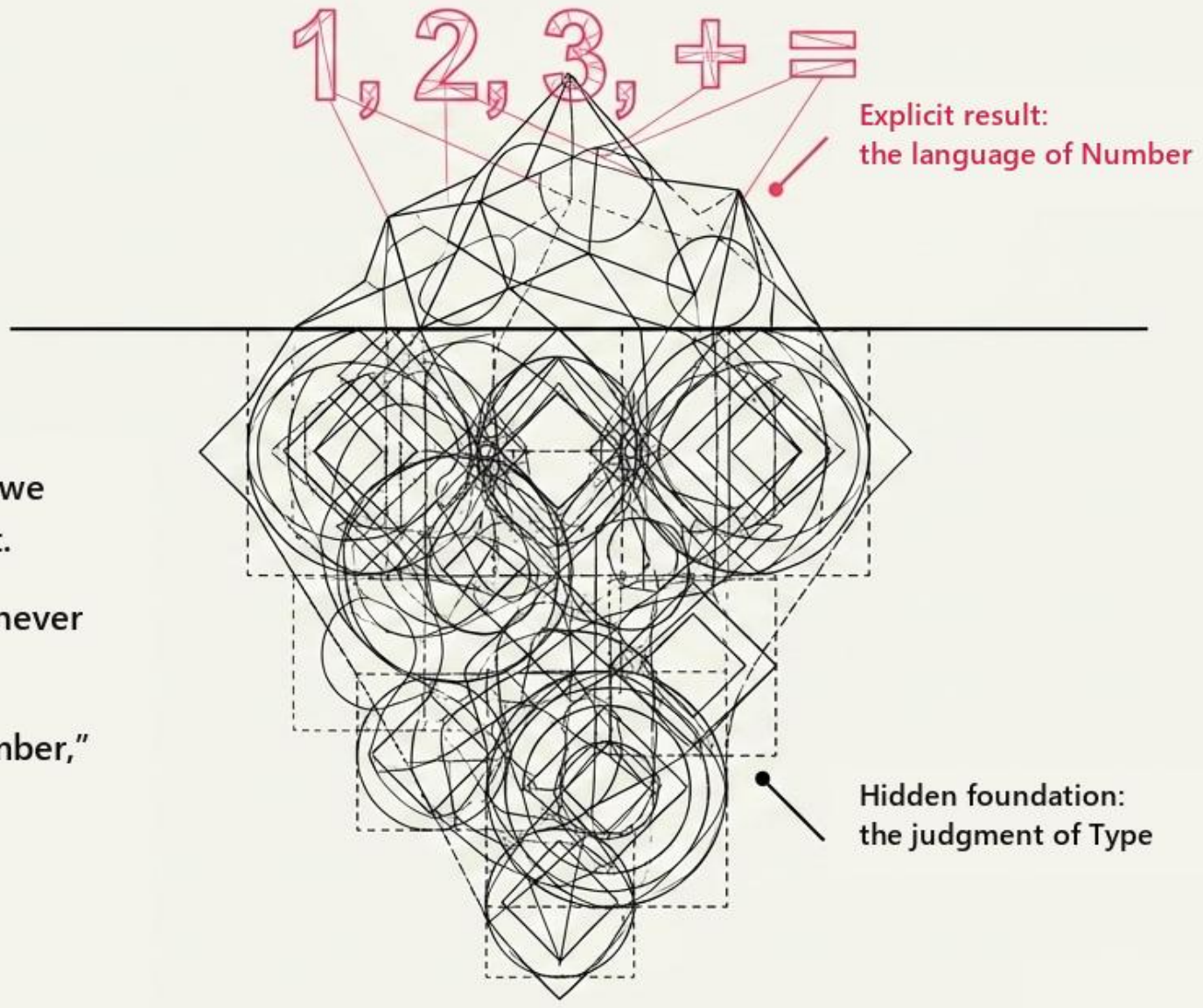
5 fruits in total

This equation only works because an unwritten step has already occurred: the brain must first judge that “apples” and “peaches” both belong to the higher category “fruit”; only then can they be treated as the same unit and added in one equation.

Apples ∈ Fruit Peaches ∈ Fruit

3 and 2 can be added because a judgment of Type has already quietly run in the background (apples, peaches → merged into fruit)—a step the mathematical symbols never show.

The Legitimacy of Mathematics Rests on the Invisible “Type”



Whether two things can be added depends on whether we can first find a “Type” and merge them into a single unit.

This step is exactly the part that mathematical notation never writes out.

Mathematics appears to be the purest language of “Number,” yet every operation depends on a vast cognitive canvas beneath it.

CONCLUSION

First Type, then Number

Type

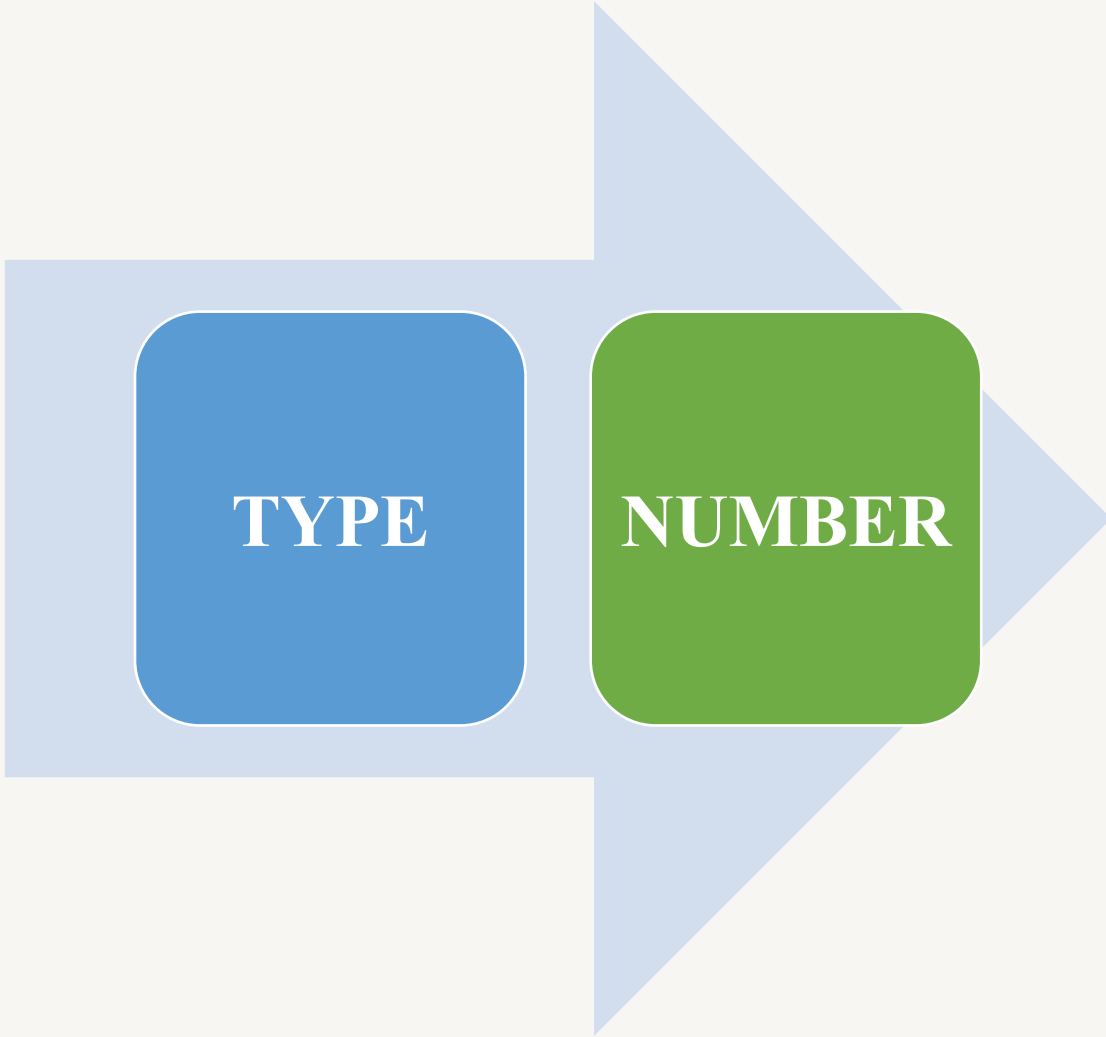
Basic Individual

The discernment of sameness and difference is the most primordial, most fundamental cognitive act

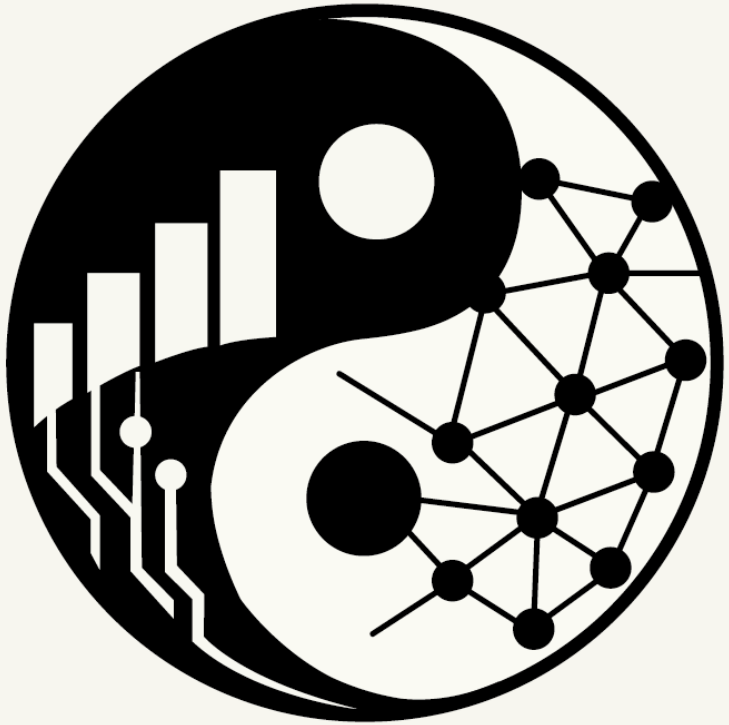
Number

Repetition of Type

On top of an already recognized Type, measure how many times it appears



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